

Experiment 15

Question

Perform Transformation using Direct Linear Transformation (DLT).

Aim

To perform image transformation using the Direct Linear Transformation (DLT) method with OpenCV.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using `cv2.imread()`.
3. Check whether the image is loaded successfully.
4. Select four source points from the original image.
5. Select four destination points.
6. Compute the transformation matrix using DLT.
7. Apply the perspective transformation.
8. Display the original and transformed images.
9. Save the transformed image.

Code

```
import cv2
```

```
import numpy as np
```

```
img = cv2.imread("dog.jpg")
```

```
rows, cols = img.shape[:2]
```

```
src = np.float32([[50,50],[300,50],[50,250],[300,250]])
```

```
dst = np.float32([[30,80],[280,30],[80,260],[260,280]])
```

```
M = cv2.getPerspectiveTransform(src, dst)
result = cv2.warpPerspective(img, M, (cols, rows))
```

```
cv2.imshow("Original", img)
```

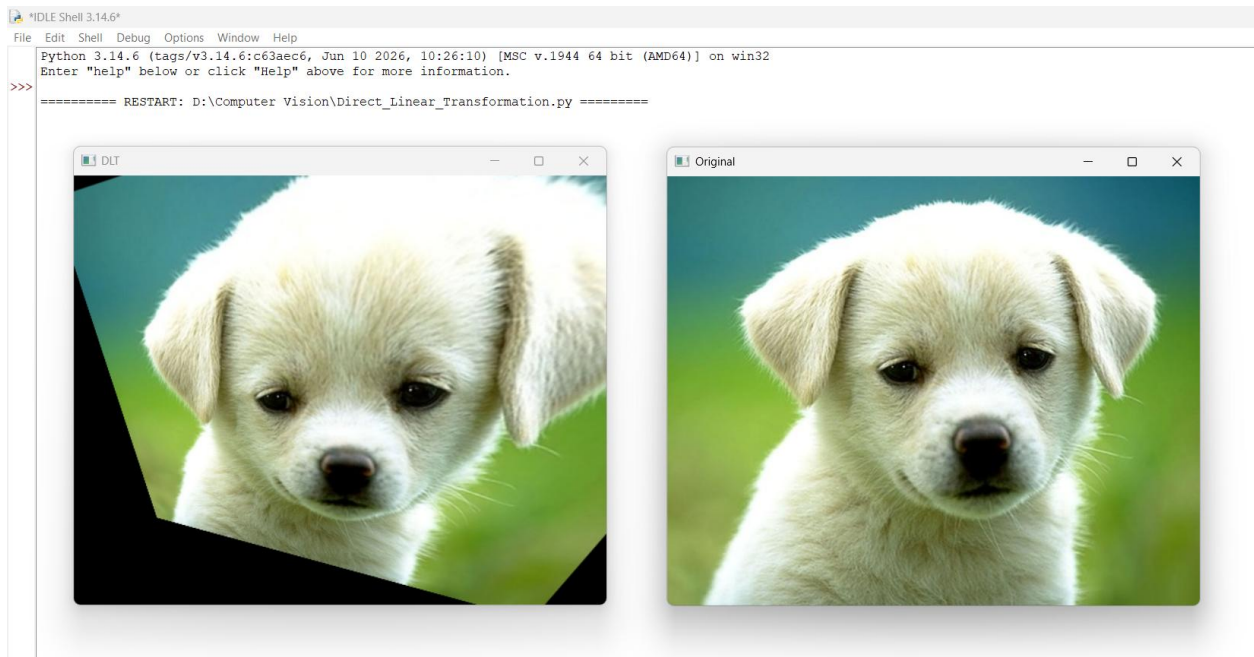
```
cv2.imshow("DLT", result)
```

Input

Input Image: dog.jpg

Output

- DLT Transformed Image



Result

The Direct Linear Transformation was successfully performed on the image using OpenCV and displayed successfully.